

Control Unit Architecture & Hardwired/Microprogrammed Control

Q. What is the role of the Control Unit? Differentiate between Hardwired Control Unit and Microprogrammed Control Unit.

> **Definition to Remember**

1. Hardwired Control Unit

The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

2. Microprogrammed (Soft-Wired) Control Unit

The control signals are generated by executing a tiny, low-level program (a **Microprogram**) stored in a special, fast memory called **Control Memory (ROM)**.

3. Key Differences

Feature	Hardwired CU	Microprogrammed CU
Control Logic	Implemented physically using logic gates, flip-flops, and decoders.	Generated by executing a microprogram stored in Control Memory (ROM).
Flexibility	Not flexible; changes require physical redesign of hardware.	Highly flexible; changes can be made by updating the microprogram.
Development Time	Longer development time due to physical design and prototyping.	Shorter development time as changes are made in software.
Cost	Higher cost due to complex hardware and prototyping.	Lower cost as it relies on standard memory and logic components.
Scalability	Difficult to scale for more complex instructions.	Easier to scale by adding more microprograms.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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Slower